MD. JAHIM UDDIN SHORIF
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EDUCATION

King Fahd University of Petroleum & Minerals Dhahran, KSA
M.S. in Civil Engineering. GPA 3.66/4.00 *[Aug. 2023-Dec. 2025]*

- Thesis title: Human Exposure and Risk Analysis of few Heavy Metals in Drinking Water: A Global Perspective.
  Supervisor: Prof. Md. Shakhawat Hossain Chowdhury, Professor, Department of CEE, KFUPM

Islamic University of Technology Gazipur, Bangladesh
B.S. in Civil Engineering. GPA 3.36/4.00 *[Jan. 2018-May 2022]*

- Thesis title: Analysis of Groundwater Quality in Dhaka Division.
  Supervisor: Dr. Md. Rezaul Karim
  Professor, Department of CEE, IUT

RESEARCH EXPERIENCES Dhahran, KSA

Supervisor: Dr. Md. Shakhawat Hossain Chowdhury

My research focuses on water and wastewater treatment, environmental health and risk assessment and computational modeling of environmental systems. My work includes probabilistic assessment of contaminants in drinking water, quantification of associated health and societal burdens, resource recovery from sewage sludge through enhanced anaerobic digestion.

- Conducted probabilistic human-health risk assessments of drinking-water contaminants, including heavy metals and N-nitrosodimethylamine (NDMA), considering multiple exposure pathways, geographical variability and uncertainty in exposure parameters.
- Monte Carlo simulation was applied to quantify contaminant exposure, carcinogenic and non-carcinogenic risks, loss of disability-adjusted life years (DALYs) and associated economic burdens. Performed sensitivity analyses to identify dominant factors and potential mitigation strategies.
- Reviewed inorganic arsenic exposure and mitigation, including its occurrence in drinking water and rice, associated human-health risks, uptake and accumulation pathways in rice plants, and potential strategies for reducing exposure.
- Applied statistical and computational modeling to climate and hydrological extremes, including spatial extreme-value modeling, intensity-duration-frequency analysis, max-stable processes and CMIP6 climate projections.

RESEARCH INTERESTS

My broad research interests are:

- Water and wastewater treatment, PFAS and heavy metals removal
- Resource recovery, anaerobic digestion, microbial electrosynthesis, CO_2 sequestration and chemical generation, waste-to-energy, bio hydrogen.
- Environmental health and risk assessment, exposure modeling
- Hydrological extremes; climate change; flood modeling and risk assessment
- Physics-informed machine learning, data-driven environmental modeling

PUBLICATIONS
(Published)

Shorif, Md Jahim Uddin, Shakhawat Chowdhury, Abdullah Al-Mamun, and Syed Masiur Rahman. "A systematic review and probabilistic meta-analysis of health risks and disability-adjusted life years (DALYs) associated with heavy metals in global drinking water: a continental comparison." *Reviews on Environmental Health* 0 (2026).

Nikmah, Norma Wihdatun, **Md Jahim Uddin Shorif**, Ahmed M. Al-Areeq, and Shakhawat Chowdhury. "Max-stable process framework for intensity–duration–frequency analysis and climate projections in monsoon Bangladesh." Geomatics, Natural Hazards and Risk 17, no. 1 (2026): 2638527.

Shorif, Md Jahim Uddin, Md Atiqur Rahman, and Shakhawat Chowdhury. "Reviews on Inorganic Arsenic in Rice Grains: Assessment of Regional Variation, Health Risk and Mitigation Strategies." Arabian Journal for Science and Engineering (2025): 1-19.

Shorif, Md Jahim Uddin, Md Atiqur Rahman, and Shakhawat Chowdhury. "Heavy metal contamination in Asian drinking water: a regional assessment of health risks and disability-adjusted life years (DALYs)." *Human and Ecological Risk Assessment: An International Journal* (2025): 1-31.

Rahman, Md Atiqur, **Md Jahim Uddin Shorif**, and Shakhawat Chowdhury. "Probabilistic Health Risk and Loss of Disability-Adjusted Life Years for N-Nitrosodimethylamine in Asian Drinking Water: Assessment and Control Strategies." *Water Environment Research* 97, no. 9 (2025): e70178.

Karim, Md Rezaul, Md Atif Arham, **Md Jahim Uddin Shorif**, Amimul Ahsan, and Nadhir Al-Ansari. "GIS based geostatistical modelling and trends analysis of groundwater quality for suitable uses in Dhaka division." *Scientific Reports* 14, no. 1 (2024): 17449.

(Submitted)

Nikmah, Norma Wihdatun, **Md Jahim Uddin Shorif**, Ahmed M. Al-Areeq, Hatim O. Sharif and Shakhawat Chowdhury. "Physics-Informed Deep Learning for Flood Inundation Modeling: Architectures, Failure Modes, and the Path Toward Hybrid Digital Twins." Reliability Engineering & System Safety.

(In Preparation)

Halima Alhimali, **Md Jahim Uddin Shorif**, Md Atiqur Rahman, Abdullah Al-Mamun, Shakhawat Chowdhury, Abdullah Al Mamun, Mohamed Shafiq Uzzaman, Jagdeep Kumar Nayak, Bashirul Haq. "Electroactive Carbon Brush for Thermophilic Anaerobic Digestion – Kinetic Performance and Economic Analysis."

Md Jahim Uddin Shorif, Md Atiqur Rahman, Shakhawat Chowdhury. "A review on Destructive Treatment and Mineralization of PFAS in Drinking Water Treatment."

(Conference)

Karim, Md Rezaul, Md Atif Arham, **Md Jahim Uddin Shorif**, and Ataur Rahman. "Analysis of groundwater quality in Dhaka division, Bangladesh." In International Conference on Water and Environmental Engineering, pp. 162-167. Science Technology and Management Crescent Australia, 2022.

WORK EXPERIENCE

Graduate Teaching Assistant, Civil and Environmental Engineering, King Fahd University of Petroleum and Minerals (KFUPM), *Dhahran, KSA* *[Aug. 2023-Present]*

TECHNICAL SKILLS

Programming language:	Python		
Engineering software:	AutoCAD 2D	GIS	ETABS
Graphics editor:	Adobe Photoshop	Adobe Illustrator	Adobe Premiere Pro
Others:	Microsoft Office		

PROFESSIONAL DEVELOPMENT

- Industrial Training Program, organized by the Department of Civil and Environmental Engineering, IUT. *[Sept. 2021-Apr. 2022]*
- Final Year Design Project, Department of Civil and Environmental Engineering, IUT. *[Oct. 2021-Apr. 2022]*
- Writing in the Sciences, Stanford Online *[Coursera 2026]*

- Programming for Everybody (Getting Started with Python), University of Michigan *[Coursera 2020]*

MEMBER AND VOLUNTEER

- Associate member of the Institution of Engineers, Bangladesh
- Lifetime member of the Islamic University of Technology Alumni Association
- Student member of the ACI IUT Student Chapter

AWARDS

- KFUPM Scholarship awarded by KFUPM to pursue my MS degree *[2023]*
- Primary Talent Pool Scholarship *[2008]*

REFERENCES

Dr. Md. Shakhawat Hossain Chowdhury Professor Department of Civil and Environmental Engineering, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia Tel: +966138602560; Fax: +966138602879 Email: Schowdhury@kfupm.edu.sa	Dr. Muhammad Shariq Vohra Professor Department of Civil and Environmental Engineering, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia Tel: +96638602854 Email: vohra@kfupm.edu.sa
Dr. Mohammad Saleh Al-Suwaiyan Professor Department of Civil and Environmental Engineering, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia Tel: +966505145439 Email: msaleh@kfupm.edu.sa	